Chapter 5. Fourier Transforms

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Overview

Results from Chapters 1–3 are cited as *Fact B.k*; they are stated in the separate *Background* notes. Fourier's idea of representing an arbitrary periodic function by a trigonometric series not only contributed greatly to solving differential equations but stimulated all of analysis, including integration theory. This chapter studies Fourier series of periodic functions and Fourier transforms of integrable functions; Fubini's theorem and convolution are the key tools.

5.1 Fourier Series

Motivation. A periodic function should be the sum of its harmonics e^{inx}. Whether and in what sense this holds is the central question; approximate identities (Fejér's kernel) give a clean positive answer for Cesàro means and show that Fourier coefficients determine the function.

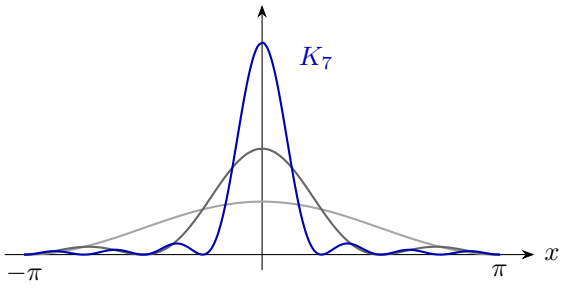


Figure. *Fejér kernels K_N ($N = 1, 3, 7$) are nonnegative, have integral 1 and concentrate at 0 as N grows. Such an approximate identity makes $K_N * f \to f$, which is why Cesàro means of Fourier series converge.*

5.1.1 The circle $\mathbb{T}$

Let $\mathbb{T}$ be the quotient of $\mathbb{R}$ by the subgroup $\{2\pi n : n = 0, \pm 1, \pm 2, \dots\}$, with the distance inherited from the real line. A 2π-periodic function can be regarded as a function on $\mathbb{T}$ and vice versa. In particular, a continuous function on $\mathbb{T}$ is a 2π-periodic continuous function, i.e. a continuous function on $[-\pi, \pi]$ with $f(-\pi) = f(\pi)$. Since $\mathbb{T} = (-\pi, \pi]$ we use Lebesgue measure, normalized to total mass 1:

$$\int_{\mathbb{T}} f = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, dx.$$

The convolution of §4.3 is defined on $L^1(\mathbb{T})$ in the same way, making $L^1(\mathbb{T})$ a Banach algebra: for $f, g : \mathbb{T} \to \mathbb{C}$, $t \in \mathbb{T}$,

$$(f * g)(t) = \int_{\mathbb{T}} f(s) g(t - s)\, ds.$$

If $f, g \in L^1(\mathbb{T})$, then as in §4.3, $(f * g)(t)$ is defined for almost every t, $f * g \in L^1(\mathbb{T})$,

$$\|f * g\|_1 \le \|f\|_1 \|g\|_1, \qquad f, g \in L^1(\mathbb{T}),$$

and the identities of Proposition 4.3.1 hold.

5.1.2 Fourier coefficients

Definition (Fourier coefficients, Fourier series)

For $f \in L^1(\mathbb{T})$ and $n = 0, \pm 1, \pm 2, \dots$ define

$$\widehat{f}(n) = \int_{\mathbb{T}} f(t) e^{-int}\, dt,$$

so that $|\widehat{f}(n)| \le \|f\|_1$. With

$$u_n(t) = e^{int}, \qquad t \in \mathbb{T}, \tag{1}$$

one checks directly that

$$u_n * f = \widehat{f}(n) u_n, \qquad n \in \mathbb{Z},\ f \in L^1(\mathbb{T}); \tag{2}$$

in particular $(u_n * f)(0) = \widehat{f}(n)$. Also $\widehat{u_n}(m) = 1$ if $m = n$ and 0 if $m \neq n$. The series

$$\sum_{n=-\infty}^{\infty} \widehat{f}(n) u_n \tag{3}$$

is the **Fourier series** of $f \in L^1(\mathbb{T})$ and $\{\widehat{f}(n) : n \in \mathbb{Z}\}$ are its **Fourier coefficients**. Without suitable conditions on the coefficients, the series (3) need not make sense as a function.

Indeed (2): $(u_n * f)(t) = \int_{\mathbb{T}} f(s) e^{in(t-s)}\, ds = e^{int} \widehat{f}(n)$.

Exercise 5.1.1

If $\widehat{f} \in \ell^1(\mathbb{Z})$, show that the series (3) converges uniformly to a continuous function.

Solution [Supplement]. $|\widehat{f}(n) u_n(t)| = |\widehat{f}(n)|$ for all t, and $\sum_n |\widehat{f}(n)| < \infty$. By the Weierstrass M-test the partial sums $\sum_{|n| \le N} \widehat{f}(n) u_n$ (or any partial sums $\sum_{-M}^{N}$) form a uniformly Cauchy sequence of continuous functions on $\mathbb{T}$; their uniform limit is continuous. □

Proposition 5.1.1

For $f, g \in L^1(\mathbb{T})$, constants $a, b \in \mathbb{C}$ and $n \in \mathbb{Z}$,

$$\widehat{af + bg}(n) = a\widehat{f}(n) + b\widehat{g}(n), \qquad \widehat{f * g}(n) = \widehat{f}(n)\, \widehat{g}(n). \tag{4}$$

Moreover for each $t \in \mathbb{T}$

$$\widehat{f_t}(n) = \widehat{f}(n) u_n(-t), \qquad n \in \mathbb{Z}, \tag{5}$$

where $f_t(s) = f(s - t)$.

Proof. The first identity follows from Fact B.10 (linearity of the integral). For the second, by Fubini,

$$\begin{aligned} \widehat{f * g}(n) &= \int_{\mathbb{T}} (f * g)(t) e^{-int}\, dt = \int_{\mathbb{T}} e^{-int} \int_{\mathbb{T}} f(t - s) g(s)\, ds\, dt \\ &= \int_{\mathbb{T}} g(s) e^{-ins} \int_{\mathbb{T}} f(t - s) e^{-in(t-s)}\, dt\, ds = \int_{\mathbb{T}} g(s) e^{-ins}\, ds \int_{\mathbb{T}} f(t) e^{-int}\, dt = \widehat{g}(n) \widehat{f}(n). \end{aligned}$$

For (5),

$$\widehat{f_t}(n) = \int_{\mathbb{T}} f(s - t) e^{-ins}\, ds = \int_{\mathbb{T}} f(s) e^{-in(s+t)}\, ds = \widehat{f}(n) u_n(-t).$$

■

Proposition 5.1.2

If f is a 2π-periodic differentiable function with $f' \in L^1(\mathbb{T})$, then

$$\widehat{f'}(n) = in\widehat{f}(n), \qquad n \in \mathbb{Z}.$$

Proof. Integrating by parts (the boundary terms cancel by periodicity),

$$\widehat{f'}(n) = \int_{\mathbb{T}} f'(t)e^{-int}\,dt = in\int_{\mathbb{T}} f(t)e^{-int}\,dt = in\widehat{f}(n).$$

■

The next theorem is a variant of Fact B.21 and can be proved with part of the method of Fact B.22.

Theorem 5.1.3

For each $p \in [1,\infty)$, the space $C(\mathbb{T})$ of continuous functions is dense in $L^p(\mathbb{T})$.

For a function f on $\mathbb{T}$ and $t \in \mathbb{T}$ define the translate as in Fact B.19:

$$f_t(s) = f(s-t), \qquad s \in \mathbb{T}.$$

The next theorem is proved like Fact B.23, using Theorem 5.1.3 instead of Fact B.22; the computation is even simpler.

Theorem 5.1.4

For each $f \in L^p(\mathbb{T})$ $(1 \le p < \infty)$, the map

$$t \mapsto f_t : \mathbb{T} \to L^p(\mathbb{T})$$

is uniformly continuous.

Exercise 5.1.2

Prove Theorems 5.1.3 and 5.1.4.

Solution [Supplement]. *Theorem 5.1.3.* Simple functions $\sum_k c_k\chi_{E_k}$ with E_k measurable are dense in $L^p(\mathbb{T})$ (as in Fact B.21), so it suffices to approximate χ_E for measurable $E \subset (-\pi,\pi)$ (the point π is a null set). Given $\epsilon > 0$, by regularity of Lebesgue measure choose a compact K and an open U with $K \subset E \subset U \subset (-\pi,\pi)$ and $m(U \setminus K) < \epsilon$. Let

$$\varphi(x) = \frac{d(x,U^c)}{d(x,U^c) + d(x,K)},$$

which is continuous on $\mathbb{R}$, $0 \le \varphi \le 1$, $\varphi = 1$ on K and $\varphi = 0$ off U. Since $U \subset (-\pi,\pi)$ is open and $K \subset U$ compact, φ vanishes near $\pm\pi$ and extends to a continuous 2π-periodic function, i.e. $\varphi \in C(\mathbb{T})$. As $|\chi_E - \varphi| \le \chi_{U\setminus K}$,

$$\|\chi_E - \varphi\|_p^p \le \frac{m(U\setminus K)}{2\pi} < \epsilon.$$

Taking linear combinations proves density.

Theorem 5.1.4. Let $f \in L^p(\mathbb{T})$ and $\epsilon > 0$; choose $g \in C(\mathbb{T})$ with $\|f-g\|_p < \epsilon$. Since $\mathbb{T}$ is compact, g is uniformly continuous: there is $\delta > 0$ with $|g(u)-g(v)| < \epsilon$ whenever $d(u,v) < \delta$. Lebesgue measure on $\mathbb{T}$ is translation invariant, so $\|f_t - g_t\|_p = \|f-g\|_p$. If $d(t,s) < \delta$, then $d(u-t,u-s) < \delta$ for every u, hence $\|g_t - g_s\|_p \le \|g_t - g_s\|_\infty \le \epsilon$ (total mass 1), and

$$\|f_t - f_s\|_p \le \|f_t - g_t\|_p + \|g_t - g_s\|_p + \|g_s - f_s\|_p < 3\epsilon.$$

□

5.1.3 Approximate identities

We are now ready for the theorem that plays the central role in the study of Fourier series. Its equation (7) says that the sequence $\langle h_n \rangle$ acts like an identity in the Banach algebra $L^1(\mathbb{T})$.

Theorem 5.1.5

If a sequence $\langle h_n \rangle$ satisfies

$$h_n \geq 0, \qquad \int_{\mathbb{T}} h_n = 1, \qquad \lim_{n\to\infty} \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n = 0 \quad (0 < \delta < \pi), \tag{6}$$

(the third condition meaning: for *every* $\delta \in (0,\pi)$ the limit is 0), then for every $f \in L^1(\mathbb{T})$

$$\lim_{n\to\infty} \|f - h_n * f\|_1 = 0. \tag{7}$$

Proof. Let $\epsilon > 0$. By Theorem 5.1.4 there is $\delta > 0$ with

$$|s| < \delta \implies \|f - f_s\|_1 < \epsilon,$$

and then N with

$$n \geq N \implies \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\, ds < \frac{\epsilon}{2\|f\|_1}.$$

For every $t \in \mathbb{T}$,

$$f(t) - (h_n * f)(t) = f(t) \int_{\mathbb{T}} h_n(s)\, ds - \int_{\mathbb{T}} h_n(s) f(t-s)\, ds = \int_{\mathbb{T}} h_n(s)\big[f(t) - f_s(t)\big] ds.$$

Hence for $n \geq N$, using Tonelli,

$$\begin{aligned}
\|f - h_n * f\|_1 &\leq \int_{\mathbb{T}} \int_{\mathbb{T}} h_n(s) |f(t) - f_s(t)|\, ds\, dt = \int_{\mathbb{T}} h_n(s) \int_{\mathbb{T}} |f(t) - f_s(t)|\, dt\, ds \\
&= \int_{\mathbb{T}} h_n(s) \|f - f_s\|_1\, ds \\
&= \int_{[-\delta,\delta]} h_n(s) \|f - f_s\|_1\, ds + \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s) \|f - f_s\|_1\, ds \\
&\leq \epsilon \int_{[-\delta,\delta]} h_n(s)\, ds + 2\|f\|_1 \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\, ds \leq \epsilon + \epsilon = 2\epsilon.
\end{aligned}$$

■

The method of proof of Theorem 5.1.5 is very typical and applies in many situations. It applies verbatim to the next theorem, where the computation is even simpler and Tonelli–Fubini is not needed.

Theorem 5.1.6

If $\langle h_n \rangle$ satisfies (6), then $\lim_{n\to\infty} \|f - h_n * f\|_\infty = 0$ for every $f \in C(\mathbb{T})$.

Proof. Since f is uniformly continuous, choose $\delta > 0$ with

$$|s| < \delta \implies \|f - f_s\|_\infty < \epsilon,$$

and N with

$$n \geq N \implies \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\, ds < \frac{\epsilon}{2\|f\|_\infty}.$$

Then for every $t \in \mathbb{T}$ and $n \geq N$,

$$|f(t) - (h_n * f)(t)| \leq \int_{\mathbb{T}} h_n(s) |f(t) - f_s(t)|\, ds = \Big(\int_{[-\delta,\delta]} + \int_{\mathbb{T}\setminus[-\delta,\delta]} \Big) h_n(s) |f(t) - f_s(t)|\, ds$$

$$\le \epsilon \int_{[-\delta,\delta]} h_n(s)\,ds + 2\|f\|_\infty \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\,ds \le \epsilon + \epsilon = 2\epsilon.$$

■

Exercise 5.1.3

In the proof of Theorem 5.1.6, show that there is $\delta > 0$ with $|s| < \delta \Rightarrow \|f - f_s\|_\infty < \epsilon$.

Solution [Supplement]. A continuous function on the compact metric space $\mathbb{T}$ is uniformly continuous: there is $\delta > 0$ with $|f(u) - f(v)| < \epsilon/2$ whenever $d(u,v) < \delta$. If $|s| < \delta$, then $d(t, t-s) \le |s| < \delta$ for every t, so $|f(t) - f_s(t)| = |f(t) - f(t-s)| < \epsilon/2$ and $\|f - f_s\|_\infty \le \epsilon/2 < \epsilon$. □

Exercise 5.1.4

Use Theorem 5.1.6 to prove Theorem 5.1.5.

Solution [Supplement]. Let $f \in L^1(\mathbb{T})$, $\epsilon > 0$, and choose $g \in C(\mathbb{T})$ with $\|f - g\|_1 < \epsilon$ (Theorem 5.1.3). Since $\mathbb{T}$ has total mass 1, $\|\cdot\|_1 \le \|\cdot\|_\infty$, and $\|h_n\|_1 = 1$, so

$$\begin{aligned}\|f - h_n * f\|_1 &\le \|f - g\|_1 + \|g - h_n * g\|_1 + \|h_n * (g - f)\|_1 \\ &\le \epsilon + \|g - h_n * g\|_\infty + \|h_n\|_1 \|g - f\|_1 < 2\epsilon + \|g - h_n * g\|_\infty.\end{aligned}$$

By Theorem 5.1.6 the last term tends to 0, so $\limsup_n \|f - h_n * f\|_1 \le 2\epsilon$. □

5.1.4 The Fejér kernel

Sequences satisfying (6) are easy to find, e.g. $\langle n\pi\chi_{[-1/n,1/n]} : n = 1, 2, \dots \rangle$. A more important example: for $n = 1, 2, \dots$ put

$$K_n = \sum_{k=-n}^{n} \Big(1 - \frac{|k|}{n+1}\Big) u_k. \tag{8}$$

$\langle K_n \rangle$ is the **Fejér kernel**.

Exercise 5.1.5

Show that $\lim_{n\to\infty} \widehat{K_n}(k) = 1$ for every $k \in \mathbb{Z}$.

Solution [Supplement]. Since $\widehat{u_j}(k) = \delta_{jk}$, $\widehat{K_n}(k) = 1 - \frac{|k|}{n+1}$ for $|k| \le n$ and 0 for $|k| > n$. For fixed k and $n \ge |k|$ this is $1 - |k|/(n+1) \to 1$. □

Since $\int_{\mathbb{T}} u_n = 0$ for $n \in \mathbb{Z} \setminus \{0\}$, the second condition $\int_{\mathbb{T}} K_n = 1$ is immediate. Expanding the left side shows

$$\Big(-\frac{1}{4}u_{-1} + \frac{1}{2} - \frac{1}{4}u_1\Big) K_n(t) = \frac{1}{n+1}\Big(-\frac{1}{4}u_{-(n+1)} + \frac{1}{2} - \frac{1}{4}u_{n+1}\Big),$$

from which

$$K_n(t) = \frac{1}{n+1} \frac{\sin^2\big(\frac{n+1}{2}t\big)}{\sin^2\big(\frac{1}{2}t\big)}, \qquad n = 0, 1, 2, \dots \tag{9}$$

Exercise 5.1.6

Prove (9).

Solution [Supplement]. For $0 \le j, l \le n$, the number of pairs with $j - l = k$ is $n + 1 - |k|$ ($|k| \le n$). Hence

$$\Big|\sum_{j=0}^{n} e^{ijt}\Big|^2 = \sum_{j,l=0}^{n} e^{i(j-l)t} = \sum_{k=-n}^{n} (n + 1 - |k|) e^{ikt} = (n+1) K_n(t).$$

For $t \neq 0$,

$$\sum_{j=0}^{n} e^{ijt} = \frac{e^{i(n+1)t} - 1}{e^{it} - 1} = \frac{e^{i(n+1)t/2}}{e^{it/2}} \cdot \frac{\sin \frac{(n+1)t}{2}}{\sin \frac{t}{2}},$$

whose squared modulus is $\sin^2 \frac{(n+1)t}{2} / \sin^2 \frac{t}{2}$. This is (9). (Equivalently, $-\frac{1}{4}u_{-1} + \frac{1}{2} - \frac{1}{4}u_1 = \frac{1}{2}(1 - \cos t) = \sin^2 \frac{t}{2}$ and similarly on the right, which is the displayed identity.) □

Since $t \mapsto \sin^2(\frac{1}{2}t)$ is increasing on $[0, \pi]$ and K_n is even, for every $\delta > 0$

$$0 \leq K_n(t) \leq \frac{1}{n+1} \frac{1}{\sin^2(\frac{1}{2}\delta)}, \qquad t \in \mathbb{T} \setminus [-\delta, \delta],$$

so $\langle K_n \rangle$ satisfies all of (6).

Exercise 5.1.7

Sketch the graph of K_n for $n = 1, 2, 3, 5, 10$.

Solution [Supplement]. $K_1(t) = 1 + \cos t$, $K_2(t) = 1 + \frac{4}{3}\cos t + \frac{2}{3}\cos 2t$, and in general K_n is even, nonnegative, with maximum $K_n(0) = n + 1$ and zeros at $t = \frac{2\pi j}{n+1}$, $j = 1, \ldots, n$. Between the zeros there are small humps of height at most $\frac{1}{(n+1)\sin^2(t/2)}$. Each K_n has average 1 over $\mathbb{T}$. As $n = 1, 2, 3, 5, 10$ the central peak grows to height $2, 3, 4, 6, 11$ while its width $\frac{4\pi}{n+1}$ shrinks, and the graph away from 0 flattens toward 0: the mass concentrates at $t = 0$. □

5.1.5 Consequences

The key property of $\langle K_n \rangle$ is that each K_n is a finite linear combination of $\{u_n : n \in \mathbb{Z}\}$. Such periodic functions are called **trigonometric polynomials**; their vector space is $T(\mathbb{T})$. The Fourier coefficients of $P = \sum_{n=-N}^{N} a_n u_n$ are $\widehat{P}(n) = a_n$.

By (2), $K_n * f$ is a trigonometric polynomial for every $f \in L^1(\mathbb{T})$, and by Theorem 5.1.5, $K_n * f \to f$ in L^1. Hence:

Theorem 5.1.7

$T(\mathbb{T})$ is dense in $L^1(\mathbb{T})$.

If $\widehat{f}(n) = 0$ for all $n \in \mathbb{Z}$, then $K_n * f = 0$ by (2), so $\|f\|_1 = \lim \|f - K_n * f\|_1 = 0$:

Theorem 5.1.8 (uniqueness)

If $f \in L^1(\mathbb{T})$ and $\widehat{f} = 0$, then $f = 0$ almost everywhere.

If $a \in \ell^1(\mathbb{Z})$, then $\sum_{k\in\mathbb{Z}} |a_k u_k(t)| = \sum_k |a_k| < \infty$ for every t, so $g(t) = \sum_{k\in\mathbb{Z}} a_k u_k(t)$ makes sense, and by Fubini (for the counting measure, Exercise 4.2.3)

$$\widehat{g}(n) = \int_{\mathbb{T}} \Big(\sum_{k\in\mathbb{Z}} a_k u_{k-n} \Big) = \sum_{k\in\mathbb{Z}} a_k \int_{\mathbb{T}} u_{k-n} = a_n.$$

In particular, if $f \in L^1(\mathbb{T})$ and $\widehat{f} \in \ell^1(\mathbb{Z})$, then $g = \sum_k \widehat{f}(k) u_k$ and f have the same Fourier coefficients, so Theorem 5.1.8 gives:

Corollary 5.1.9

If $f \in L^1(\mathbb{T})$ and $\widehat{f} \in \ell^1(\mathbb{Z})$, then

$$f(t) = \sum_{n\in\mathbb{Z}} \widehat{f}(n) e^{int}$$

for almost every $t \in \mathbb{T}$.

On the other hand, given $\epsilon > 0$ take $P \in T(\mathbb{T})$ with $\|f - P\|_1 < \epsilon$ and let $N = \max\{|n| : \widehat{P}(n) \neq 0\}$. For $|n| > N$,

$$|\widehat{f}(n)| = |\widehat{f}(n) - \widehat{P}(n)| \leq \|f - P\|_1 < \epsilon.$$

This proves:

Theorem 5.1.10 (Riemann–Lebesgue)

$\lim_{n\to\pm\infty} \widehat{f}(n) = 0$ for every $f \in L^1(\mathbb{T})$.

Exercise 5.1.8

If a 2π-periodic function $f : \mathbb{R} \to \mathbb{C}$ is twice differentiable and $f'' \in L^1(\mathbb{T})$, show that $\widehat{f} \in \ell^1(\mathbb{Z})$.

Solution [Supplement]. f' is differentiable, hence continuous and so bounded on $\mathbb{T}$; thus $f' \in L^1(\mathbb{T})$. Applying Proposition 5.1.2 to f and to f',

$$\widehat{f''}(n) = in\,\widehat{f'}(n) = (in)^2 \widehat{f}(n) = -n^2 \widehat{f}(n).$$

Hence $|\widehat{f}(n)| = |\widehat{f''}(n)|/n^2 \leq \|f''\|_1/n^2$ for $n \neq 0$, and $\sum_n |\widehat{f}(n)| \leq |\widehat{f}(0)| + \|f''\|_1 \sum_{n\neq 0} n^{-2} < \infty$. □

Let $c_0(\mathbb{Z})$ be the vector space of functions $a : \mathbb{Z} \to \mathbb{C}$ with $\lim_{n\to\pm\infty} a(n) = 0$, with the norm of $\ell^\infty(\mathbb{Z})$.

Exercise 5.1.9

Show that the normed space $c_0(\mathbb{Z})$ is a Banach space.

Solution [Supplement]. $\ell^\infty(\mathbb{Z})$ is complete, so it suffices to show $c_0(\mathbb{Z})$ is closed in it. Let $a_k \in c_0(\mathbb{Z})$, $\|a_k - a\|_\infty \to 0$, and $\epsilon > 0$. Choose k with $\|a_k - a\|_\infty < \epsilon$, then N with $|a_k(n)| < \epsilon$ for $|n| > N$. For $|n| > N$, $|a(n)| \leq |a(n) - a_k(n)| + |a_k(n)| < 2\epsilon$. So $a \in c_0(\mathbb{Z})$. □

Summarizing:

Theorem 5.1.11

The map $f \mapsto \widehat{f}$ is a norm-decreasing, injective linear map from the Banach algebra $L^1(\mathbb{T})$ into $c_0(\mathbb{Z})$.

Proof. Linearity is (4); $\|\widehat{f}\|_\infty \leq \|f\|_1$ is the estimate $|\widehat{f}(n)| \leq \|f\|_1$; $\widehat{f} \in c_0(\mathbb{Z})$ is Theorem 5.1.10; injectivity is Theorem 5.1.8. ■

With the pointwise product $(ab)(n) = a(n)b(n)$, $c_0(\mathbb{Z})$ is a Banach algebra. The last identity of (4) says that $f \mapsto \widehat{f}$ preserves multiplication between $L^1(\mathbb{T})$ and $c_0(\mathbb{Z})$. A linear map between Banach algebras that preserves multiplication is a **homomorphism**.

Remark ($L^1(\mathbb{T})$ has no identity)

If the Banach algebra $L^1(\mathbb{T})$ had an identity e, then for every $n \in \mathbb{Z}$

$$\widehat{e}(n) = \widehat{e}(n)\widehat{u_n}(n) = \widehat{e * u_n}(n) = \widehat{u_n}(n) = 1,$$

contradicting $\widehat{e} \in c_0(\mathbb{Z})$. So $L^1(\mathbb{T})$ has no identity; Theorem 5.1.5 (an *approximate identity*) is the device that overcomes this.

Exercise 5.1.10

If $\langle h_n \rangle$ satisfies (6), show that $\|a\widehat{h_n} - a\|_\infty \to 0$ for every $a \in c_0(\mathbb{Z})$.

Solution [Supplement]. First, $|\widehat{h_n}(m)| \le \|h_n\|_1 = 1$. Next, for fixed m and $0 < \delta < \pi$,

$$|1 - \widehat{h_n}(m)| = \Big| \int_{\mathbb{T}} h_n(s)(1 - e^{-ims})\, ds \Big| \le \int_{[-\delta,\delta]} h_n(s)|ms|\, ds + 2\int_{\mathbb{T}\setminus[-\delta,\delta]} h_n \le |m|\delta + 2\int_{\mathbb{T}\setminus[-\delta,\delta]} h_n,$$

so $\limsup_n |1 - \widehat{h_n}(m)| \le |m|\delta$ for every δ, i.e. $\widehat{h_n}(m) \to 1$. Now let $\epsilon > 0$ and choose M with $|a(m)| < \epsilon$ for $|m| > M$. Then

$$\|a\widehat{h_n} - a\|_\infty \le \max_{|m|\le M} |a(m)||\widehat{h_n}(m) - 1| + \sup_{|m|>M} |a(m)| \cdot 2,$$

where the first term tends to 0 (finitely many m) and the second is at most 2ϵ. □

Finally, consider some consequences of Theorem 5.1.6. With the Fejér kernel, $K_n * f \in T(\mathbb{T})$ and $K_n * f \to f$ uniformly for $f \in C(\mathbb{T})$:

Corollary 5.1.12

$T(\mathbb{T})$ is dense in $C(\mathbb{T})$ (in the uniform norm).

If $f \in L^p(\mathbb{T})$ $(1 \le p < \infty)$, Theorem 5.1.3 gives $g \in C(\mathbb{T})$ with $\|f - g\|_p < \epsilon$, and Corollary 5.1.12 gives $P \in T(\mathbb{T})$ with $\|g - P\|_\infty < \epsilon$. Then

$$\|f - P\|_p \le \|f - g\|_p + \|g - P\|_p < \epsilon + \|g - P\|_\infty < 2\epsilon.$$

Corollary 5.1.13

$T(\mathbb{T})$ is dense in $L^p(\mathbb{T})$ for $1 \le p < \infty$.

Corollary 5.1.12 closely resembles the Weierstrass theorem that a continuous function on a bounded closed interval can be uniformly approximated by polynomials, and in fact implies it.

Theorem 5.1.14 (Weierstrass)

The space of polynomial functions on $[0, 1]$ is dense in $C[0, 1]$.

Proof. Extend a continuous $f : [0, 1] \to \mathbb{C}$ to a continuous periodic $g : \mathbb{T} \to \mathbb{C}$. By Corollary 5.1.12 choose $\{a_k \in \mathbb{C} : k = 0, \pm 1, \ldots, \pm n\}$ with

$$\Big| f(t) - \sum_{k=-n}^{n} a_k u_k(t) \Big| < \epsilon, \qquad t \in [0, 1].$$

Since the Taylor series of the exponential converges uniformly on $[0,1]$, for each k there is $M(k) \in \mathbb{N}$ with

$$\Big| \sum_{\ell=0}^{M(k)} \frac{(ikt)^\ell}{\ell!} - u_k(t) \Big| \le \frac{\epsilon}{(2n+1)|a_k|+1}, \qquad t \in [0,1].$$

Put

$$P(t) = \sum_{k=-n}^{n} a_k \sum_{\ell=0}^{M(k)} \frac{(ikt)^\ell}{\ell!}, \qquad t \in [0,1].$$

Then $|f(t) - P(t)| < 2\epsilon$ (Exercise 5.1.11). ■

Exercise 5.1.11

Complete the proof of Theorem 5.1.14.

Solution [Supplement]. *Extension.* Since $[0,1] \subset (-\pi,\pi)$, define g on $[-\pi,\pi]$ by $g = f$ on $[0,1]$, g linear from $f(1)$ at $t=1$ to 0 at $t=2$, $g=0$ on $[2,\pi] \cup [-\pi,-1]$, and g linear from 0 at $t=-1$ to $f(0)$ at $t=0$. Then g is continuous with $g(-\pi) = g(\pi)$, so $g \in C(\mathbb{T})$.

Estimate. P is a polynomial in t. For $t \in [0,1]$,

$$\begin{aligned} |f(t) - P(t)| &\le \Big| f(t) - \sum_{k=-n}^{n} a_k u_k(t) \Big| + \sum_{k=-n}^{n} |a_k| \Big| u_k(t) - \sum_{\ell=0}^{M(k)} \frac{(ikt)^\ell}{\ell!} \Big| \\ &< \epsilon + \sum_{k=-n}^{n} \frac{|a_k|\epsilon}{(2n+1)|a_k|+1} < \epsilon + (2n+1) \cdot \frac{\epsilon}{2n+1} = 2\epsilon. \end{aligned}$$

(If f is real-valued, $\operatorname{Re} P$ is a real polynomial with $|f - \operatorname{Re} P| \le |f - P|$.) □

5.2 The Fourier Integral

Motivation. For nonperiodic functions on $\mathbb{R}$ the sum over frequencies becomes an integral. We transfer the series theory to the transform: continuity and decay of $\hat{f}$, inversion via Gauss and Fejér kernels, and the rules turning differentiation into multiplication.

5.2.1 Definition and continuity

Definition (Fourier transform)

For $f \in L^1(\mathbb{R})$ and $\alpha \in \mathbb{R}$ define

$$\widehat{f}(\alpha) = \int_{\mathbb{R}} f(x) e^{-i\alpha x}\, dx.$$

$\widehat{f}$ is the **Fourier transform** of f.

Clearly $|\widehat{f}(\alpha)| \le \|f\|_1$ for all $\alpha \in \mathbb{R}$. If $\alpha_n \to \alpha$, then

$$|\widehat{f}(\alpha_n) - \widehat{f}(\alpha)| \le \int_{\mathbb{R}} |f(x)| |e^{-i\alpha_n x} - e^{-i\alpha x}|\, dx,$$

and since $|e^{-i\alpha_n x} - e^{-i\alpha x}| \le 2$ and $e^{-i\alpha_n x} \to e^{-i\alpha x}$, the Lebesgue convergence theorem gives $\widehat{f}(\alpha_n) \to \widehat{f}(\alpha)$. So $\widehat{f}$ *is always a bounded continuous function.* Writing

$$u_\alpha(x) = e^{i\alpha x}, \qquad x \in \mathbb{R},$$

one checks immediately that

$$u_\alpha * f = \widehat{f}(\alpha) u_\alpha, \qquad \alpha \in \mathbb{R}.$$

Example 1

For $f(x) = e^{-|x|/a}$ $(a > 0)$,

$$\widehat{f}(\alpha) = \int_{\mathbb{R}} e^{-|x|/a} e^{-i\alpha x}\, dx = \int_{-\infty}^{0} e^{(1/a - i\alpha)x}\, dx + \int_{0}^{\infty} e^{(-1/a - i\alpha)x}\, dx = \frac{2a}{1 + a^2\alpha^2}.$$

For $g = \chi_{[a,b]}$,

$$\widehat{g}(\alpha) = \int_{-\infty}^{\infty} \chi_{[a,b]} e^{-i\alpha x}\, dx = \frac{e^{-i\alpha a} - e^{-i\alpha b}}{i\alpha},$$

and in particular

$$\widehat{\chi_{[-a,a]}}(\alpha) = \frac{2\sin \alpha a}{\alpha}, \qquad a > 0.$$

Exercise 5.2.1

In Example 1, describe how the graphs of f and $\widehat{f}$ change as $a \to \infty$. Do the same for $\widehat{\chi_{[-a,a]}}$.

Solution [Supplement]. As $a \to \infty$, $f(x) = e^{-|x|/a}$ spreads out and increases to the constant 1. Its transform $\widehat{f}(\alpha) = \frac{2a}{1+a^2\alpha^2}$ is a Cauchy-type bump with peak $2a$ at $\alpha = 0$ and half-width $1/a$; it becomes taller and narrower while keeping total integral $\int \widehat{f} = 2\pi f(0) = 2\pi$, i.e. it concentrates at $\alpha = 0$ (like $2\pi\delta_0$). Similarly $\widehat{\chi_{[-a,a]}}(\alpha) = \frac{2\sin a\alpha}{\alpha}$ has central peak $2a$, zeros at $\alpha = k\pi/a$ $(k \neq 0)$, and oscillations decaying like $2/|\alpha|$. As $a \to \infty$ the peak grows, the zeros crowd together and the oscillation becomes faster: the transform of a wide function is a narrow one. (Note $\frac{2\sin a\alpha}{\alpha} \notin L^1(\mathbb{R})$.) □

The basic properties of Proposition 5.1.1 hold for Fourier integrals as well.

Proposition 5.2.1

For $f, g \in L^1(\mathbb{R})$ and constants $a, b \in \mathbb{C}$,

$$\widehat{af + bg}(\alpha) = a\widehat{f}(\alpha) + b\widehat{g}(\alpha), \qquad \widehat{f * g}(\alpha) = \widehat{f}(\alpha)\,\widehat{g}(\alpha), \qquad \alpha \in \mathbb{R}.$$

Moreover for each $\alpha \in \mathbb{R}$

$$\widehat{fu_\alpha} = (\widehat{f})_\alpha, \qquad \widehat{f_x}(\alpha) = \widehat{f}(\alpha) u_\alpha(-x), \quad x \in \mathbb{R}.$$

Proof. For example, for every $\beta \in \mathbb{R}$

$$\widehat{fu_\alpha}(\beta) = \int_{\mathbb{R}} f(x) u_\alpha(x) e^{-i\beta x}\, dx = \int_{\mathbb{R}} f(x) e^{-i(\beta-\alpha)x}\, dx = \widehat{f}(\beta - \alpha),$$

so $\widehat{fu_\alpha} = (\widehat{f})_\alpha$. The rest is Exercise 5.2.2. ■

Exercise 5.2.2

Complete the proof of Proposition 5.2.1.

Solution [Supplement]. Linearity follows from linearity of the integral. For the convolution, $(x, y) \mapsto f(x - y)g(y)e^{-i\alpha x}$ is integrable on $\mathbb{R}^2$ (its absolute value integrates to $\|f\|_1\|g\|_1$), so by

Fubini

$$\widehat{f * g}(\alpha) = \int_{\mathbb{R}} g(y) e^{-i\alpha y} \int_{\mathbb{R}} f(x-y) e^{-i\alpha(x-y)}\, dx\, dy = \widehat{g}(\alpha)\widehat{f}(\alpha).$$

For translation, $\widehat{f_x}(\alpha) = \int_{\mathbb{R}} f(y-x)e^{-i\alpha y}\, dy = \int_{\mathbb{R}} f(y)e^{-i\alpha(y+x)}\, dy = \widehat{f}(\alpha)e^{-i\alpha x} = \widehat{f}(\alpha)u_\alpha(-x)$. □

5.2.2 Differentiation

Proposition 5.2.2

Let $f \in L^1(\mathbb{R})$.

(a) Let $g(x) = -ixf(x)$. If $g \in L^1(\mathbb{R})$, then $\widehat{f}$ is differentiable and

$$\widehat{f}\,'(\alpha) = \widehat{g}(\alpha), \qquad \alpha \in \mathbb{R}.$$

(b) If f is differentiable and $f' \in L^1(\mathbb{R})$, then

$$\widehat{f'}(\alpha) = i\alpha\widehat{f}(\alpha), \qquad \alpha \in \mathbb{R}.$$

Proof. (a) The difference quotient is

$$\frac{\widehat{f}(\alpha+h) - \widehat{f}(\alpha)}{h} = \int_{\mathbb{R}} f(x)e^{-i\alpha x}\frac{e^{-ihx}-1}{h}\, dx,$$

and

$$\left| f(x)e^{-i\alpha x}\frac{e^{-ihx}-1}{h} \right| \le |xf(x)|.$$

Since $x \mapsto |xf(x)|$ is integrable and $\lim_{h\to 0}\frac{e^{-ihx}-1}{h} = -ix$, the Lebesgue convergence theorem gives, for every real sequence $h_n \to 0$,

$$\lim_{n\to\infty} \frac{\widehat{f}(\alpha+h_n) - \widehat{f}(\alpha)}{h_n} = \widehat{g}(\alpha).$$

(b) As in Proposition 5.1.2, integrate by parts: $\widehat{f'}(\alpha) = \int_{\mathbb{R}} f'(x)e^{-i\alpha x}\, dx = i\alpha \int_{\mathbb{R}} f(x)e^{-i\alpha x}\, dx$; the boundary terms vanish because $f(x) \to 0$ as $x \to \pm\infty$ (f has limits at $\pm\infty$ since $f' \in L^1$, and they must be 0 since $f \in L^1$). ■

5.2.3 Riemann–Lebesgue lemma

As for Fourier series, $\lim_{\alpha\to\pm\infty}\widehat{f}(\alpha) = 0$ for $f \in L^1(\mathbb{R})$. Using $e^{\pi i} = -1$,

$$\widehat{f}(\alpha) = -\int_{\mathbb{R}} f(x)e^{-i\alpha(x+\pi/\alpha)}\, dx = -\int_{\mathbb{R}} f\Big(x - \frac{\pi}{\alpha}\Big)e^{-i\alpha x}\, dx,$$

and hence

$$2\widehat{f}(\alpha) = \int_{\mathbb{R}} [f(x) - f_{\pi/\alpha}(x)]e^{-i\alpha x}\, dx.$$

By Fact B.23,

$$0 \le \lim_{\alpha\to\pm\infty}|\widehat{f}(\alpha)| \le \frac{1}{2}\lim_{\alpha\to\pm\infty}\|f - f_{\pi/\alpha}\|_1 = 0.$$

Since $\widehat{f}$ is already known to be continuous:

Theorem 5.2.3 (Riemann–Lebesgue)

$\widehat{f} \in C_0(\mathbb{R})$ for every $f \in L^1(\mathbb{R})$.

5.2.4 Approximate identities on $\mathbb{R}$

Using Fact B.23 instead of Theorem 5.1.4 in the proof of Theorem 5.1.5 yields:

Theorem 5.2.4

If a family $\{h_\lambda : \lambda \in (0,\infty)\}$ satisfies

$$h_\lambda \geq 0, \qquad \int_{\mathbb{R}} h_\lambda = 1, \qquad \lim_{\lambda\to\infty} \int_{\mathbb{R}\setminus[-\delta,\delta]} h_\lambda = 0 \quad (\delta > 0), \tag{10}$$

then:

(a) $\lim_{\lambda\to\infty} \|f - h_\lambda * f\|_1 = 0$ for every $f \in L^1(\mathbb{R})$;

(b) $\lim_{\lambda\to\infty} \|f - h_\lambda * f\|_\infty = 0$ for every bounded uniformly continuous $f : \mathbb{R} \to \mathbb{C}$.

Exercise 5.2.3

Prove Theorem 5.2.4.

Solution [Supplement]. (a) Let $\epsilon > 0$. By Fact B.23 choose $\delta > 0$ with $|s| < \delta \Rightarrow \|f - f_s\|_1 < \epsilon$, and Λ with $\int_{\mathbb{R}\setminus[-\delta,\delta]} h_\lambda < \epsilon/(2\|f\|_1)$ for $\lambda \geq \Lambda$. Since $f(t) - (h_\lambda * f)(t) = \int_{\mathbb{R}} h_\lambda(s)[f(t) - f_s(t)]\,ds$, Tonelli gives for $\lambda \geq \Lambda$

$$\|f - h_\lambda * f\|_1 \leq \int_{\mathbb{R}} h_\lambda(s)\|f - f_s\|_1\,ds \leq \epsilon \int_{[-\delta,\delta]} h_\lambda + 2\|f\|_1 \int_{\mathbb{R}\setminus[-\delta,\delta]} h_\lambda < 2\epsilon.$$

(b) Choose δ with $|s| < \delta \Rightarrow \|f - f_s\|_\infty < \epsilon$ (uniform continuity) and Λ with $\int_{\mathbb{R}\setminus[-\delta,\delta]} h_\lambda < \epsilon/(2\|f\|_\infty)$ for $\lambda \geq \Lambda$. Then for every t,

$$|f(t) - (h_\lambda * f)(t)| \leq \int_{\mathbb{R}} h_\lambda(s)|f(t) - f_s(t)|\,ds \leq \epsilon + 2\|f\|_\infty \cdot \frac{\epsilon}{2\|f\|_\infty} = 2\epsilon.$$

□

If $h : \mathbb{R} \to \mathbb{C}$ satisfies

$$h \geq 0, \qquad \int_{\mathbb{R}} h = 1, \tag{11}$$

put

$$h_\lambda(x) = \lambda h(\lambda x), \qquad x \in \mathbb{R},\ \lambda > 0. \tag{12}$$

Then $\{h_\lambda : \lambda > 0\}$ satisfies (10): $h_\lambda \geq 0$ and $\int h_\lambda = 1$ are clear, and for fixed $\delta > 0$, as $\lambda \to \infty$,

$$\int_{\mathbb{R}\setminus[-\delta,\delta]} h_\lambda(x)\,dx = \int_{\mathbb{R}\setminus[-\lambda\delta,\lambda\delta]} h(x)\,dx \to 0.$$

Exercise 5.2.4

Show that $\widehat{h_\lambda}(\alpha) = \widehat{h}\left(\frac{\alpha}{\lambda}\right)$ for every $\alpha \in \mathbb{R}$.

Solution [Supplement]. Substituting $y = \lambda x$: $\widehat{h_\lambda}(\alpha) = \int_{\mathbb{R}} \lambda h(\lambda x) e^{-i\alpha x}\,dx = \int_{\mathbb{R}} h(y) e^{-i(\alpha/\lambda)y}\,dy = \widehat{h}(\alpha/\lambda)$. □

The case where h satisfying (11) has the form

$$h(x) = \frac{1}{2\pi}\int_{\mathbb{R}} p(\alpha)u_\alpha(x)\,d\alpha \tag{13}$$

is very useful. Here $p:\mathbb{R}\to\mathbb{C}$ is assumed to satisfy

$$0\le p\le 1, \qquad p(0)=\lim_{\alpha\to 0}p(\alpha)=1, \qquad p\in L^1(\mathbb{R}). \tag{14}$$

A simple example satisfying all requirements:

$$H(x)=\frac{1}{2\pi}\int_{\mathbb{R}} e^{-|\alpha|}e^{i\alpha x}\,d\alpha = \frac{1}{\pi}\frac{1}{1+x^2}, \tag{15}$$

and then

$$H_\lambda(x)=\frac{1}{2\pi}\int_{\mathbb{R}} e^{-|\alpha/\lambda|}e^{i\alpha x}\,d\alpha = \frac{1}{2\pi}\frac{2\lambda}{1+\lambda^2x^2}.$$

$\{H_\lambda:\lambda>0\}$ is the **Poisson kernel**.

5.2.5 Smooth functions are dense

Let $C_c^n(\mathbb{R})$ be the functions in $C_c(\mathbb{R})$ that are n times differentiable with continuous $f^{(n)}$, and $C_c^\infty(\mathbb{R})=\bigcap_{n=1}^\infty C_c^n(\mathbb{R})$. If $h\in C_c^\infty(\mathbb{R})$ satisfies (11), then $h_\lambda\in C_c^\infty(\mathbb{R})$, and by repeated use of Proposition 4.3.2, $h_\lambda * f\in C_c^\infty(\mathbb{R})$ for every $f\in C_c(\mathbb{R})$. Moreover by Theorem 5.2.4

$$\lim_{\lambda\to\infty}\|f-h_\lambda*f\|_1=\lim_{\lambda\to\infty}\|f-h_\lambda*f\|_\infty=0.$$

Since $C_c(\mathbb{R})$ is dense in $L^1(\mathbb{R})$ and in $C_0(\mathbb{R})$:

Proposition 5.2.5

$C_c^\infty(\mathbb{R})$ is dense in the normed spaces $L^1(\mathbb{R})$ and $C_0(\mathbb{R})$.

Exercise 5.2.5

Give an example of $h\in C_c^\infty(\mathbb{R})$ satisfying (11).

Solution [Supplement]. Let $\varphi(x)=\exp(-\frac{1}{1-x^2})$ for $|x|<1$ and $\varphi(x)=0$ for $|x|\ge 1$. Since $\psi(t)=e^{-1/t}$ $(t>0)$, $\psi(t)=0$ $(t\le 0)$ is C^∞ (every derivative on $t>0$ is $P(1/t)e^{-1/t}$ for a polynomial P, which tends to 0 as $t\downarrow 0$), and $\varphi(x)=\psi(1-x^2)$, $\varphi\in C_c^\infty(\mathbb{R})$, $\varphi\ge 0$, $\int\varphi>0$. Take $h=\varphi/\int_{\mathbb{R}}\varphi$. □

5.2.6 The inversion formula

For h given by (13) and $\{h_\lambda\}$ by (12), compute $h_\lambda * f$ for $f\in L^1(\mathbb{R})$ using Fubini:

$$\begin{aligned}(h_\lambda*f)(x) &= \int_{\mathbb{R}} f(x-y)\frac{\lambda}{2\pi}\int_{\mathbb{R}} p(\alpha)e^{i\alpha\lambda y}\,d\alpha\,dy = \frac{1}{2\pi}\int_{\mathbb{R}} f(x-y)\int_{\mathbb{R}} p(\tfrac{\alpha}{\lambda})e^{i\alpha y}\,d\alpha\,dy\\ &= \frac{1}{2\pi}\int_{\mathbb{R}} p(\tfrac{\alpha}{\lambda})\int_{\mathbb{R}} f(x-y)e^{i\alpha y}\,dy\,d\alpha = \frac{1}{2\pi}\int_{\mathbb{R}} p(\tfrac{\alpha}{\lambda})\int_{\mathbb{R}} f(y)e^{i\alpha(x-y)}\,dy\,d\alpha\\ &= \frac{1}{2\pi}\int_{\mathbb{R}} p(\tfrac{\alpha}{\lambda})\widehat{f}(\alpha)e^{i\alpha x}\,d\alpha.\end{aligned}$$

(Fubini applies since $\iint|f(x-y)||p(\alpha/\lambda)|\,d\alpha\,dy=\lambda\|f\|_1\|p\|_1<\infty$.)

From now on assume p, h satisfy (14), (11) and (13). Then

$$(h_\lambda * f)(x) = \frac{1}{2\pi}\int_{\mathbb{R}} p\Big(\frac{\alpha}{\lambda}\Big)\widehat{f}(\alpha)e^{i\alpha x}\,d\alpha.$$

Since

$$\Big|p\Big(\frac{\alpha}{\lambda}\Big)\widehat{f}(\alpha)e^{i\alpha x}\Big| \le |\widehat{f}(\alpha)|, \qquad \lim_{\lambda\to\infty} p\Big(\frac{\alpha}{\lambda}\Big) = 1,$$

if $\widehat{f} \in L^1(\mathbb{R})$ the Lebesgue convergence theorem gives

$$\lim_{\lambda\to\infty}(h_\lambda * f)(x) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{f}(\alpha)e^{i\alpha x}\,d\alpha, \qquad x \in \mathbb{R}.$$

On the other hand $\lim_{\lambda\to\infty}\|h_\lambda * f - f\|_1 = 0$ by Theorem 5.2.4, so by Fact B.18 there is a sequence $\lambda_n \to \infty$ with

$$\lim_{n\to\infty}(h_{\lambda_n} * f)(x) = f(x)$$

for almost every x. This proves the analogue of Corollary 5.1.9 (with, e.g., $p(\alpha) = e^{-|\alpha|}$ and $h = H$ from (15)):

Theorem 5.2.6 (inversion formula)

If $f \in L^1(\mathbb{R})$ and $\widehat{f} \in L^1(\mathbb{R})$, then

$$f(x) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{f}(\alpha)e^{i\alpha x}\,d\alpha \tag{16}$$

for almost every x.

The right side of (16) is a continuous function of x (as in the continuity of $\widehat{f}$). Hence if f is continuous, (16) holds for every $x \in \mathbb{R}$.

Example 2

Applying the inversion formula to Example 1 (with $a = 1$),

$$e^{-|x|} = \int_{-\infty}^{\infty}\frac{1}{\pi(1+\alpha^2)}e^{i\alpha x}\,d\alpha = \int_{-\infty}^{\infty}\frac{\cos\alpha x}{\pi(1+\alpha^2)}\,d\alpha + i\int_{-\infty}^{\infty}\frac{\sin\alpha x}{\pi(1+\alpha^2)}\,d\alpha.$$

Since the left side is real (and the cosine integrand is even),

$$\int_0^{\infty}\frac{\cos\alpha x}{1+\alpha^2}\,d\alpha = \frac{\pi}{2}e^{-|x|}.$$

If $\widehat{f} = 0$, then $\widehat{f} \in L^1$ and (16) gives $f = 0$ almost everywhere, the analogue of Theorem 5.1.8:

Theorem 5.2.7 (uniqueness)

If $f \in L^1(\mathbb{R})$ and $\widehat{f} = 0$, then $f = 0$ almost everywhere.

5.2.7 Fejér and Gauss kernels

Two more examples of p satisfying (14), (11), (13). Recalling the definition (8) of the Fejér kernel, for $\lambda > 0$ put

$$K_\lambda(x) = \frac{1}{2\pi}\int_{\mathbb{R}}\max\Big\{1-\frac{|\alpha|}{\lambda},0\Big\}u_\alpha(x)\,d\alpha = \frac{1}{2\pi}\int_{-\lambda}^{\lambda}\Big(1-\frac{|\alpha|}{\lambda}\Big)e^{i\alpha x}\,d\alpha.$$

Then

$$\lambda K_1(\lambda x) = \frac{1}{2\pi}\int_{-1}^{1}(1-|\alpha|)e^{i\alpha\lambda x}\lambda\,d\alpha = K_\lambda(x),$$

so with $\lambda = 1$ in mind define

$$K(x) = \frac{1}{2\pi}\int_{-1}^{1}(1-|\alpha|)e^{i\alpha x}\,d\alpha.$$

A direct computation (integration by parts) gives

$$K_\lambda(x) = \frac{1}{\pi}\int_0^\lambda \Big(1-\frac{\alpha}{\lambda}\Big)\cos\alpha x\,d\alpha = \frac{1}{\pi\lambda x^2}(1-\cos\lambda x) = \frac{1}{2\pi\lambda}\,\frac{\sin^2(\frac{\lambda x}{2})}{(\frac{x}{2})^2},$$

so $K \geq 0$. Replacing α by $-\alpha$ in the defining integral of K_λ does not change it, so with $f_\lambda(\alpha) = \max\{1-\frac{|\alpha|}{\lambda}, 0\}$ we have $K_\lambda = \frac{1}{2\pi}\widehat{f_\lambda}$. Since f_λ is integrable and $\widehat{f_\lambda} = 2\pi K_\lambda = O(x^{-2})$ is integrable, the inversion formula gives

$$\int_{-\infty}^{\infty} K_\lambda(x)e^{i\alpha x}\,dx = \frac{1}{2\pi}\int_{-\infty}^{\infty}\widehat{f_\lambda}(x)e^{i\alpha x}\,dx = f_\lambda(\alpha) = \max\Big\{1-\frac{|\alpha|}{\lambda},0\Big\}.$$

Since f_λ is continuous, this holds for every $\alpha\in\mathbb{R}$; in particular for $\lambda = 1$, $\alpha = 0$,

$$\int_{-\infty}^{\infty} K(x)\,dx = 1.$$

The family $\{K_\lambda : \lambda > 0\}$ obtained this way is also called the **Fejér kernel**.

Finally let

$$G(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-\alpha^2/2}u_\alpha(x)\,d\alpha.$$

By Fact B.11 (differentiation under the integral) and integration by parts,

$$G'(x) = \frac{1}{2\pi}\int_{\mathbb{R}} i\alpha e^{-\alpha^2/2}e^{i\alpha x}\,d\alpha = -xG(x)$$

(indeed $\int i\alpha e^{-\alpha^2/2}e^{i\alpha x}\,d\alpha = -i\int(e^{-\alpha^2/2})'e^{i\alpha x}\,d\alpha = i\cdot ix\int e^{-\alpha^2/2}e^{i\alpha x}\,d\alpha$). Since $G(0) = \frac{1}{2\pi}\sqrt{2\pi} = \frac{1}{\sqrt{2\pi}}$, solving the ODE gives

$$G(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2},$$

so $G\geq 0$ and $\int_{\mathbb{R}} G = 1$. The family $\{G_\lambda : \lambda > 0\}$ is the **Gauss kernel**.

5.2.8 Transforms of the kernels

The inversion formula says that if g and $\widehat{g}$ are continuous L^1-functions, then

$$g(x) = \frac{1}{2\pi}\widehat{\widehat{g}}(-x), \qquad x\in\mathbb{R}. \tag{17}$$

Suppose $f, g\in L^1(\mathbb{R})$, g continuous, satisfy

$$f(x) = \frac{1}{2\pi}\int_{\mathbb{R}} g(\alpha)e^{i\alpha x}\,d\alpha, \qquad x\in\mathbb{R}.$$

Then $f(x) = \frac{1}{2\pi}\widehat{g}(-x)$, so $\widehat{g}\in L^1(\mathbb{R})$. By the inversion formula for g (valid everywhere, g being continuous),

$$g(y) = \frac{1}{2\pi}\int_{\mathbb{R}}\widehat{g}(\alpha)e^{i\alpha y}\,d\alpha = \int_{\mathbb{R}} f(-\alpha)e^{i\alpha y}\,d\alpha = \int_{\mathbb{R}} f(\beta)e^{-i\beta y}\,d\beta = \widehat{f}(y).$$

So $\widehat{f} = g$. Applying this to the three kernels:

$$\widehat{K_\lambda}(\alpha) = \max\Big\{1-\frac{|\alpha|}{\lambda},0\Big\}, \qquad \widehat{P_\lambda}(\alpha) = e^{-|\alpha/\lambda|}, \qquad \widehat{G_\lambda}(\alpha) = e^{-\alpha^2/2\lambda^2},$$

where $P_\lambda = H_\lambda$ is the Poisson kernel.

Exercise 5.2.6

For $f \in L^1(\mathbb{R})$ express $\widehat{K_\lambda * f}$ in terms of $\widehat{f}$. Use this to show that $\{f \in L^1(\mathbb{R}) : \widehat{f} \in C_c(\mathbb{R})\}$ is dense in $L^1(\mathbb{R})$, and compare with Theorem 5.1.7 for Fourier series.

Solution [Supplement]**.** By Proposition 5.2.1,

$$\widehat{K_\lambda * f}(\alpha) = \widehat{K_\lambda}(\alpha)\widehat{f}(\alpha) = \max\left\{1 - \frac{|\alpha|}{\lambda}, 0\right\}\widehat{f}(\alpha),$$

which is continuous and vanishes outside $[-\lambda, \lambda]$; so $K_\lambda * f$ belongs to the set in question. Since $K_\lambda(x) = \lambda K(\lambda x)$ with $K \geq 0$, $\int K = 1$, the family satisfies (10), and Theorem 5.2.4 gives $\|K_\lambda * f - f\|_1 \to 0$. Hence the set is dense. Comparison: on $\mathbb{T}$, $\widehat{K_n * f}(k) = (1 - \frac{|k|}{n+1})\widehat{f}(k)$ vanishes for $|k| > n$, i.e. $K_n * f$ is a trigonometric polynomial. "Compactly supported Fourier transform" on $\mathbb{R}$ is the analogue of "finitely many nonzero Fourier coefficients" (trigonometric polynomials) on $\mathbb{T}$, and both density statements are proved by the same Fejér-kernel argument. □

5.2.9 The Fourier transform as a homomorphism

By now we know that $f \mapsto \widehat{f}$ is an injective linear map from $L^1(\mathbb{R})$ into $C_0(\mathbb{R})$. If $f \in C_c^2(\mathbb{R})$, then $f'' \in L^1(\mathbb{R})$ and by Proposition 5.2.2(b) $\widehat{f''}(\alpha) = -\alpha^2\widehat{f}(\alpha)$; since $\widehat{f''}$ is bounded and $\widehat{f}$ is bounded and continuous, $|\widehat{f}(\alpha)| \leq \min\{\|f\|_1, \|f''\|_1/\alpha^2\}$, so $\widehat{f} \in L^1(\mathbb{R})$. By (17), f is the Fourier transform of another L^1-function ($f(x) = \widehat{k}(x)$ with $k(\alpha) = \frac{1}{2\pi}\widehat{f}(-\alpha)$). Thus $C_c^2(\mathbb{R})$ lies in the range of $f \mapsto \widehat{f} : L^1(\mathbb{R}) \to C_0(\mathbb{R})$, and $C_c^2(\mathbb{R}) \supset C_c^\infty(\mathbb{R})$ is dense in $C_0(\mathbb{R})$ by Proposition 5.2.5:

Theorem 5.2.8

The Fourier transform $f \mapsto \widehat{f}$ is an injective, norm-decreasing homomorphism from the Banach algebra $L^1(\mathbb{R})$ into $C_0(\mathbb{R})$, and its range is dense.

Here the multiplication on $C_0(\mathbb{R})$ is the pointwise product $(fg)(x) = f(x)g(x)$. The map $f \mapsto \widehat{f} : L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ of Theorem 5.1.11 is also called a Fourier transform in this broader sense.

Exercise 5.2.7

Show that the range of $f \mapsto \widehat{f} : L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ is dense.

Solution [Supplement]**.** Every finitely supported $a : \mathbb{Z} \to \mathbb{C}$ is the transform of the trigonometric polynomial $\sum_n a(n)u_n \in L^1(\mathbb{T})$. For $a \in c_0(\mathbb{Z})$ let $a_N = a\chi_{[-N,N]}$; then $\|a - a_N\|_\infty = \sup_{|n|>N}|a(n)| \to 0$. So finitely supported sequences, all in the range, are dense in $c_0(\mathbb{Z})$. □

5.2.10 Poisson summation

How are Fourier series and Fourier integrals related? Given $f \in L^1(\mathbb{R})$, as in Fact B.12 the series

$$F(t) = \sum_{n=-\infty}^{\infty} f(t - 2n\pi) \tag{18}$$

converges absolutely for almost every $t \in \mathbb{T}$, and $F \in L^1(\mathbb{T})$ (indeed $\int_{\mathbb{T}} \sum_n |f(t-2n\pi)|\,dt = \frac{1}{2\pi}\|f\|_1$ by term-by-term integration). By Levi's theorem (term-by-term integration),

$$\widehat{F}(n) = \int_{\mathbb{T}} F(t)e^{-int}\,dt = \sum_{k=-\infty}^{\infty} \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t + 2k\pi)e^{-int}\,dt = \frac{1}{2\pi}\int_{\mathbb{R}} f(t)e^{-int}\,dt = \frac{1}{2\pi}\widehat{f}(n). \tag{19}$$

Hence if the Fourier series of F converges at $t \in \mathbb{T}$ (to $F(t)$), we get the **Poisson summation formula**

$$\sum_{n=-\infty}^{\infty} f(t-2n\pi) = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} \widehat{f}(n) e^{int}.$$

Exercise 5.2.8

Use $f(x) = e^{-|x|}$ to prove

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2+1} = \pi \, \frac{e^{2\pi}+1}{e^{2\pi}-1}.$$

Solution [Supplement]. Here $\widehat{f}(\alpha) = \frac{2}{1+\alpha^2}$ (Example 1), so $\widehat{F}(n) = \frac{1}{\pi}\frac{1}{1+n^2}$ and $\widehat{F} \in \ell^1(\mathbb{Z})$. The series $F(t) = \sum_n e^{-|t-2n\pi|}$ converges uniformly on $[-\pi,\pi]$ (for $|n| \ge 1$ its terms are at most $e^{-(2|n|-1)\pi}$), so F is continuous. By Corollary 5.1.9 and Exercise 5.1.1, $F(t) = \sum_n \widehat{F}(n) e^{int}$ a.e., both sides continuous, hence for every t. At $t = 0$:

$$\sum_n e^{-2\pi|n|} = \frac{1}{\pi} \sum_n \frac{1}{1+n^2}.$$

The left side is $1 + \frac{2e^{-2\pi}}{1-e^{-2\pi}} = \frac{1+e^{-2\pi}}{1-e^{-2\pi}} = \frac{e^{2\pi}+1}{e^{2\pi}-1}$, which gives the formula. □

Exercise 5.2.9

For $f \in L^1(\mathbb{R})$ and $\lambda > 0$ define, as in (18),

$$F_\lambda(t) = \sum_{n=-\infty}^{\infty} f_\lambda(t-2n\pi), \qquad f_\lambda(x) = \lambda f(\lambda x).$$

Show that

$$\widehat{F_\lambda}(n) = \frac{1}{2\pi} \widehat{f}\Big(\frac{n}{\lambda}\Big), \qquad \lim_{\lambda\to\infty} \|F_\lambda\|_{L^1(\mathbb{T})} = \frac{1}{2\pi}\|f\|_{L^1(\mathbb{R})}.$$

Use this to prove Theorem 5.2.7.

Solution [Supplement]. By (19) and Exercise 5.2.4, $\widehat{F_\lambda}(n) = \frac{1}{2\pi}\widehat{f_\lambda}(n) = \frac{1}{2\pi}\widehat{f}(n/\lambda)$. Upper bound: $\|F_\lambda\|_{L^1(\mathbb{T})} \le \frac{1}{2\pi}\sum_n \int_{-\pi}^{\pi} |f_\lambda(t-2n\pi)|\,dt = \frac{1}{2\pi}\|f_\lambda\|_1 = \frac{1}{2\pi}\|f\|_1$. Lower bound: separating the $n=0$ term,

$$\|F_\lambda\|_{L^1(\mathbb{T})} \ge \frac{1}{2\pi}\Big[\int_{-\pi}^{\pi} |f_\lambda(t)|\,dt - \sum_{n\ne 0}\int_{-\pi}^{\pi} |f_\lambda(t-2n\pi)|\,dt\Big] = \frac{1}{2\pi}\Big[\int_{|x|\le\pi} |f_\lambda| - \int_{|x|>\pi} |f_\lambda|\Big]$$
$$= \frac{1}{2\pi}\Big[\int_{|y|\le\lambda\pi} |f| - \int_{|y|>\lambda\pi} |f|\Big] \longrightarrow \frac{1}{2\pi}\|f\|_1 \quad (\lambda\to\infty).$$

Proof of Theorem 5.2.7: if $\widehat{f} = 0$, then $\widehat{F_\lambda}(n) = 0$ for all n and λ, so $F_\lambda = 0$ a.e. by Theorem 5.1.8, i.e. $\|F_\lambda\|_{L^1(\mathbb{T})} = 0$. Letting $\lambda \to \infty$ gives $\|f\|_1 = 0$. □